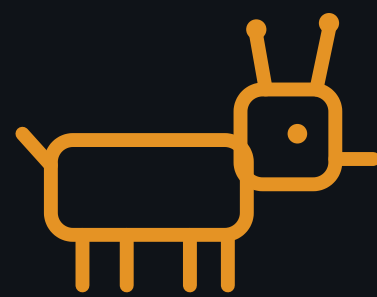


Kai 9



A 3D-printed companion robot that helps older adults live independently, cleaning the home, keeping a quiet eye on wellbeing, and turning everyday messages into a friendly spoken voice.

Imperial College London

Electrical & Electronic Engineering · Group Design Project

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6
ENGINEERED SUBSYSTEMS

2× Pi
PI 5 COMPUTE · PI 4 SENSORY

4
POWER-PROTECTION LAYERS

12.8 V
LiFePO4 · SELF-CHARGING

IBM Bob
ACROSS THE WHOLE BUILD

BUS · SOURCE · POWER & DISTRIBUTION

One protected 12.8 V bus feeds the whole robot

A single LiFePO4 battery powers every load through a **fused distribution bus**, with each branch independently protected and the pack guarded by four safety layers. When idle, Kai 9 drives onto a dock and recharges itself.

Protection **internal BMS** → **30 A main fuse** → **master kill** → **branch fuses**

Charging **base station** → **pogo pins** → **Ø60 mm chassis plates (+ top / – bottom)** → **15 A fuse** → **battery+**

Direct negative return; robust 16 AWG wiring handles the 10 A maximum charger output.

BATTERY Azeco **12.8 V 20 Ah** LiFePO4 · internal BMS

DISTRIBUTION BlueSea **6-way** fused bus · 30 A main + master kill

F1 · 5A Pi 5 + RPLiDAR (buck → **5.1 V** rail)

F2 · 20A Cytron **MDD10A** motor driver

F3 · 10A Vacuum · logic-level **MOSFET** (GPIO)

F4 · 5A ZK-502C audio amplifier

INFRASTRUCTURE · DUAL-PI TOPOLOGY

Distributed processing architecture

System responsibilities are split across a dual-processor array linked via high-speed **ZMQ over Ethernet**. The **Raspberry Pi 4** acts as the sensory edge, hosting Whisperflow and TTS to control microphone, speaker, and camera interactions. Central execution lives on a **Raspberry Pi 5** running a fully functional **LangGraph agent powered by Llama 3**.

Llama 3 LLM LangGraph Agent Framework ZMQ Communication Layer Whisperflow Speech-to-Text
TTS Audio Pipeline Ethernet Backhaul

LOAD · MOTION

Autonomous navigation

Kai 9 maps each room as it moves and plans its own way around furniture and people. A spinning laser scanner and wheel odometry feed a ROS 2 stack that turns routes into precise wheel commands.

ROS 2 Jazzy · Pi 5 RPLiDAR A1 · 360° wheel-encoder odometry Nav2 + SLAM Toolbox MDD10A drive
live obstacle avoidance

LOAD · PERCEPTION

On-sensor AI camera

A dedicated AI camera recognises objects on the sensor itself, then hands the computer ready-made positions and labels — so heavy image processing never touches the navigation brain.

Sony IMX500 on-sensor inference boxes · classes · confidence runs on companion Pi 4 keeps Pi 5 free

BODY · FRAME

Chassis & structure

A rigid internal frame carries the heavy parts — battery and vacuum — while a rounded 3D-printed shell gives Kai 9 its approachable, dog-like form: no sharp edges, a compact body that slips past furniture, a neutral grey finish.

load-bearing internal frame 3D-printed shell vacuum built in as the tail battery = compute deck
rounded · edge-free · grey

LINK · WHATSAPP

Digital communication gateway

Family reach the user through ordinary WhatsApp messages. Kai 9 reads them aloud, summarises long ones, and answers simple questions like weather, status or what's on today. **Only approved contacts get through**, and every message is checked before it reaches the robot.

WhatsApp Cloud API webhook two FastAPI services authorised-sender filter status · weather · calendar
spoken summaries

CARE · WELLBEING

AI companion & wellbeing

Kai 9 keeps a gentle eye on the daily routine and offers nudges to move, stretch or call a loved one — always encouragement, never instruction. If prompts go unanswered, it can quietly escalate to a carer or family member.

presence sensing · mmWave activity & social prompts carer escalation non-patronising dialogue privacy-first · local

MVP AT THE BOOTH

Wheeled locomotion Obstacle avoidance Vacuum cleaning WhatsApp gateway Wellbeing & presence check Activity prompts IBM Bob voice Self-charging dock

INTELLIGENCE LAYER · IBM BOB & AGENT CORE

System Integration, accelerated by IBM Bob

IBM Bob served as our ultimate engineering co-pilot throughout the development cycle, vastly speeding up our Python code assembly, architectural reasoning, and technical documentation. By deploying Bob as our primary intelligence framework, we easily optimised our LangGraph tool execution and structured asynchronous memory routines, allowing the robot to communicate with natural dialogue while running efficiently within tight compute limits.

DEVELOPMENT CO-PILOT

Accelerated Python scripting, system debugging, and documentation pipelines.

DETERMINISTIC SANDBOX

Enforces type-safe argument validation between Llama 3 and the hardware layer. The agent is strictly limited to pre-defined Python functions with hardcoded boundary caps, preventing unauthorised motor or vacuum overrides regardless of model hallucinations or prompt injection attempts.

DATA SECURITY & SILOING

Secures user privacy through local message sanitisation on the sensory edge. The LangGraph state memory completely decouples raw WhatsApp payloads and care calendars from external network exposure, operating entirely within restricted local data structures.

ASYNCHRONOUS MEMORY

Organises logs and processes long-term context during idle states.

Every prompt and system output is logged in bob_log.md to feed our validation report. All AI output is reviewed and corrected by the team before it reaches the user.